A Project Report on

IoT-BASED SMART HOME AUTOMATION WITH MOBILE MONITORING

Submitted in partial fulfilment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering with Artificial Intelligence and Machine Learning

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This is to certify that the report entitled "**IoT-Based Smart Home Automation with Mobile Monitoring**" is prepared and submitted by **OHM SHRIVASTAVA (Reg. No. 25BAI1057)** to Vellore Institute of Technology, Chennai, in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering with Artificial Intelligence and Machine Learning** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

The Internet of Things (IoT) has emerged as a transformative technology enabling seamless integration of physical devices with digital intelligence. This project presents the design and implementation of an IoT-based Smart Room Hub System using the ESP8266 NodeMCU microcontroller—a cost-effective, WiFi-enabled platform suitable for embedded automation applications. The system addresses key limitations of conventional home automation systems, particularly the reliance on passive infrared (PIR) motion sensors that can only detect movement rather than sustained occupancy.

The Smart Room Hub employs an infrared (IR) obstacle sensor for continuous presence detection, a DHT11 sensor for real-time temperature and humidity monitoring, a 16×2 I2C liquid crystal display (LCD) for on-site status visualization, and two dual-channel relay modules controlling four LED indicators that provide intuitive visual feedback. A hierarchical control algorithm ensures energy efficiency: temperature-based indicators activate only when a person is detected, preventing unnecessary relay switching in unoccupied rooms.

The system features a built-in ESP8266WebServer framework for future remote monitoring via a web browser, making it extensible to cloud platforms such as ThingSpeak or Blynk. Testing confirmed accurate presence detection with a 10-second presence timeout, reliable DHT11 readings within ±2°C and ±5% RH tolerances, responsive LCD updates every two seconds, and correct conditional relay logic at the 25°C temperature threshold. The project demonstrates practical proficiency in embedded C++ programming, I2C communication, relay-based load control, and IoT system architecture—skills directly applicable to smart home, industrial automation, and building management domains.

Keywords: IoT, ESP8266, NodeMCU, Smart Room Hub, IR obstacle sensor, DHT11, relay control, home automation, WiFi, environmental monitoring.

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Place: Chennai

Date: **OHM SHRIVASTAVA**

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LIST OF ABBREVIATIONS

ADC - Analogue-to-Digital Converter
DHT - Digital Humidity and Temperature
GPIO - General-Purpose Input/Output
HTTP - Hypertext Transfer Protocol
HVAC - Heating, Ventilation, and Air Conditioning
I2C - Inter-Integrated Circuit
IoT - Internet of Things
IR - Infrared
LCD - Liquid Crystal Display
MQTT - Message Queuing Telemetry Transport
PIR - Passive Infrared
PWM - Pulse-Width Modulation
RH - Relative Humidity
RTC - Real-Time Clock
SDA - Serial Data
SCL - Serial Clock
SoC - System-on-Chip
VCC - Voltage at the Common Collector (supply voltage)
WPA2 - Wi-Fi Protected Access 2

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The proliferation of low-cost, WiFi-capable microcontrollers has democratized the design of Internet of Things (IoT) devices, enabling students, hobbyists, and engineers alike to prototype sophisticated automation systems with modest resources. Among these platforms, the Espressif ESP8266 SoC—popularized in the NodeMCU development board—stands out for its dual role as a fully featured microcontroller and a WiFi radio transceiver, eliminating the need for a separate networking module.

Conventional home automation systems typically rely on passive infrared (PIR) sensors for occupancy detection. While inexpensive and widely available, PIR sensors are inherently movement-triggered: a person seated quietly at a desk will cause the sensor to register absence after a few seconds of inactivity, often resulting in lights or HVAC systems switching off unexpectedly. This fundamental limitation motivates the use of alternative sensing modalities that provide persistent presence detection.

Simultaneously, environmental monitoring—specifically temperature and humidity— has become a routine feature of smart room systems, offering data that drives comfort management and energy conservation decisions. The DHT11 sensor, though modest in accuracy, provides a cost-effective solution for simultaneous temperature and humidity sensing within a single digital interface.

1.2 PROBLEM STATEMENT

Existing entry-level home automation projects suffer from three recurring deficiencies:

- Transient presence detection: PIR sensors fail to maintain a sustained occupancy signal for stationary occupants, leading to premature deactivation of room services.

- Isolated sensing: Temperature monitoring systems typically operate independently of occupancy logic, wasting energy by conditioning vacant rooms.
- Limited feedback: Status information is often confined to the serial monitor, inaccessible to non-technical end-users without a connected computer.

This project addresses all three gaps through a unified, hierarchically controlled IoT hub that couples IR-based continuous presence detection with occupancy-conditional temperature monitoring and a local LCD display.

1.3 OBJECTIVES

The primary objectives of this project are as follows:

1. Design and implement a continuous room occupancy detection subsystem using an IR obstacle sensor with a configurable 10-second timeout.
2. Integrate real-time temperature and humidity sensing using the DHT11 sensor with conditional activation linked to room occupancy state.
3. Provide immediate visual status feedback via four LED indicators (RED and GREEN pairs) governed by relay modules, implementing a hierarchical control logic.
4. Display environmental data and occupancy status on a 16×2 I2C LCD panel, updated at 2-second intervals.
5. Establish a WiFi-connected web server framework for future remote monitoring and control capabilities.

1.4 SCOPE OF THE PROJECT

The scope of this project encompasses hardware design, embedded firmware development, and functional validation of the Smart Room Hub system on a breadboard prototype. Cloud integration, persistent data logging, and mobile application development are identified as future extensions but are not within the deliverables of the current implementation. The system is designed for single-room deployment, though a multi-room architecture using MQTT is described as an expansion pathway.

1.5 MAPPING TO SUSTAINABLE DEVELOPMENT GOALS

This project aligns with the United Nations Sustainable Development Goal 7 (Affordable and Clean Energy) by implementing occupancy-conditional control that prevents unnecessary energy consumption in vacant rooms. It additionally supports SDG 9 (Industry, Innovation, and Infrastructure) by demonstrating accessible IoT innovation using low-cost, globally available components, and SDG 11 (Sustainable Cities and Communities) through the application of smart building principles at the residential scale.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The design of the Smart Room Hub draws on a substantial body of work spanning IoT architectures, occupancy detection methodologies, environmental monitoring systems, and relay-based embedded control. This chapter surveys relevant literature and establishes the theoretical framework within which the present work is situated.

2.2 IOT ARCHITECTURES FOR SMART HOMES

Gubbi et al. (2013) defined the IoT as an interconnected infrastructure of physical objects embedded with sensors, actuators, and network connectivity that enables data collection and actuation without requiring direct human intervention. Their vision has since materialized in commercial products and academic prototypes alike. The ESP8266—a SoC released by Espressif Systems—represents a pivotal enabler of lowcost IoT development, integrating a Tensilica L106 RISC processor with a full 802.11 b/g/n WiFi stack (Espressif Systems, 2022).

Kodali and Mahesh (2016) demonstrated the utility of the NodeMCU platform for home automation, showing that the Arduino IDE's cross-compilation toolchain could be leveraged to develop firmware for ESP8266-based systems with minimal overhead. Their work confirmed WiFi reliability for intra-home communication at distances representative of typical residential room layouts.

2.3 OCCUPANCY DETECTION TECHNOLOGIES

The choice of occupancy sensor significantly influences system performance. PIR sensors are the most commonly employed technology in entry-level automation systems due to their low cost and zero-standby-power operation. However, Dodier et al. (2006) found that PIR-based occupancy models exhibited false-negative rates exceeding 30% for sedentary occupants in office settings, a finding corroborated by Labeodan et al. (2015) in a comprehensive review of building occupancy detection technologies.

IR obstacle sensors—reflective infrared proximity detectors—offer an alternative paradigm: rather than detecting body heat movement, they emit a modulated IR beam and measure reflectance. This provides a static presence signal for any object within the configured detection range, making them inherently immune to the inactivity blind spots of PIR sensors. The trade-off is reduced detection range (typically 2–30 cm versus 5–7 m for PIR sensors), positioning the IR sensor as appropriate for close-proximity desktop or workstation monitoring scenarios.

Ultrasonic distance sensors (e.g., HC-SR04) represent a medium-range alternative with detection distances of 2–400 cm (Eltek Ltd., 2019), while computer vision approaches using camera-based depth sensing achieve room-level occupancy maps at the cost of significantly higher computational requirements and privacy concerns.

2.4 ENVIRONMENTAL SENSING

The DHT11 is a widely documented sensor in IoT education literature. ElectronicWings (2019) provided a comprehensive guide to its single-wire digital communication protocol, noting that the sensor's 1 Hz sampling rate and ±2°C / ±5% RH accuracy are sufficient for comfort monitoring but inadequate for precision industrial applications. For higher accuracy requirements, the DHT22 (±0.5°C, ±2–5% RH) or SHT31 (±0.3°C, ±2% RH via I2C) are recommended alternatives. The present project employs the DHT11 in recognition of its cost and availability advantages for a student-level implementation.

2.5 RELAY-BASED LOAD CONTROL

Opto-isolated relay modules are the canonical method for interfacing low-voltage microcontroller GPIO pins with higher-voltage or higher-current loads. Horowitz and Hill (2015) detail the electrical rationale: the opto-isolator decouples the microcontroller's 3.3 V logic domain from the relay coil's 5 V drive circuit, and the relay's mechanical contacts from the load circuit entirely, providing both logic-level protection and potential for AC mains load switching. The active-LOW triggering convention adopted by the relay modules used in this project (GPIO LOW = relay energized) provides an inherent fail-safe: in the event of a microcontroller reset, all GPIO pins default HIGH, placing all relays in the off state.

2.6 WEB-BASED IOT DASHBOARDS

The ESP8266WebServer library enables the NodeMCU to host lightweight HTTP endpoints directly, eliminating the need for an external server for basic monitoring tasks. Instructables (2020) documented a similar DHT11-over-ESP8266 web dashboard, demonstrating round-trip latency below 200 ms over a local area network. For cloud-extended deployments, Blynk and ThingSpeak provide managed platforms with established ESP8266 client libraries and mobile front-ends (ThingSpeak, 2023; Blynk Inc., 2023).

2.7 I2C LCD INTEGRATION ON ESP8266

The Arduino forum thread documented by Community (2021) specifically addresses the non-default I2C pin assignment on ESP8266-based boards. The standard Arduino Wire library maps SDA to pin A4 and SCL to pin A5 on AVR-based boards; however, on ESP8266 NodeMCU, the Wire.begin(SDA, SCL) overload must be explicitly called with the desired GPIO numbers (D2 = GPIO4 for SDA, D1 = GPIO5 for SCL in this project) to initialize the I2C bus on the correct physical pins. Failure to invoke this overload results in the LCD remaining unresponsive, a common pitfall in student projects.

2.8 SUMMARY

The literature establishes that IR obstacle sensors are a suitable alternative to PIR sensors for close-proximity continuous presence detection; DHT11 sensors are adequate for educational-grade environmental monitoring; opto-isolated active-LOW relay modules provide safe embedded load control; and ESP8266 NodeMCU's integrated WiFi and extensive library ecosystem make it an appropriate platform for entry-level IoT automation systems. The Smart Room Hub system synthesizes these proven subsystems into a coherent, hierarchically controlled whole.

CHAPTER 3

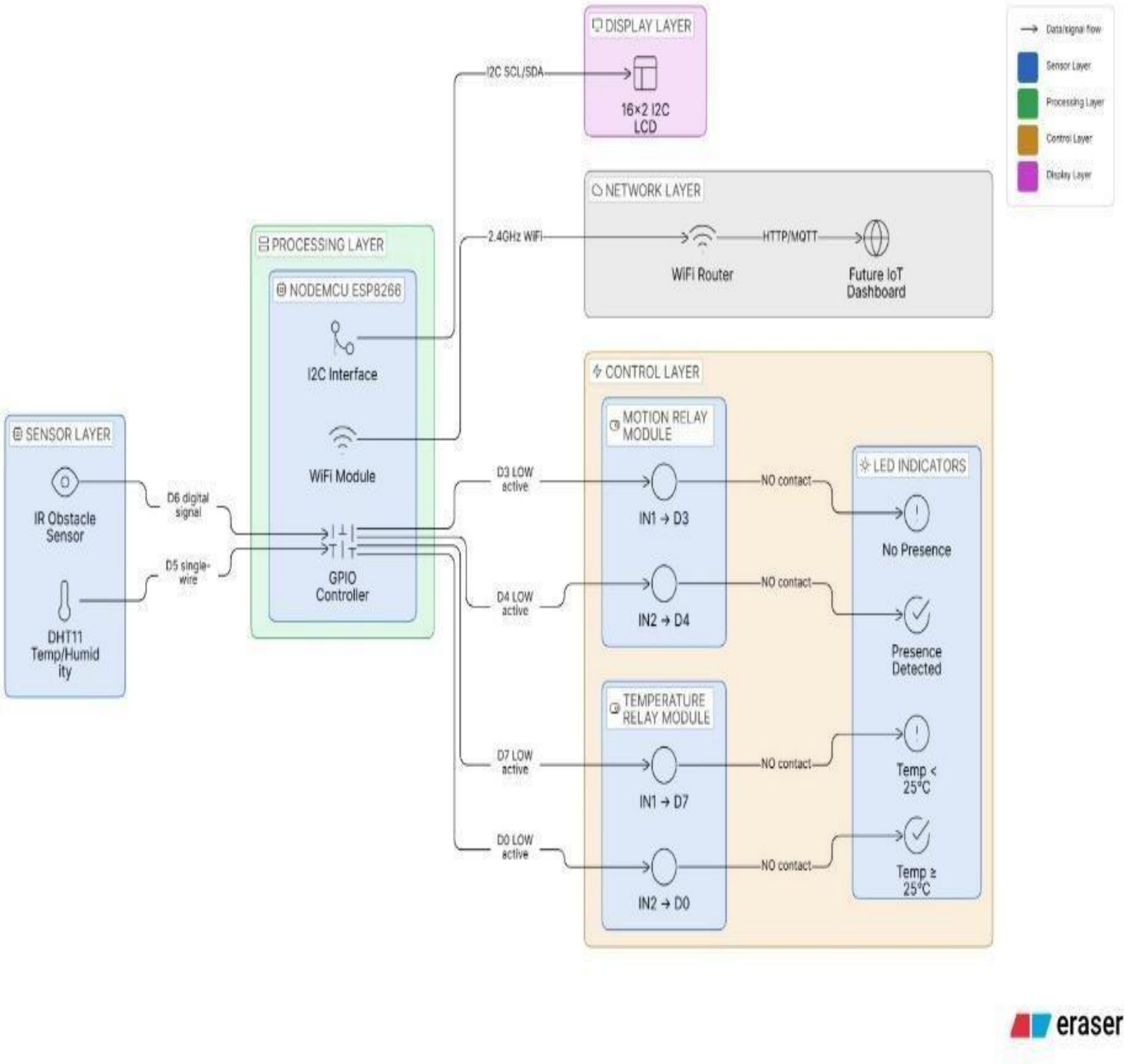
METHODOLOGY

3.1 SYSTEM ARCHITECTURE

The Smart Room Hub adopts a single-board, sensor-actuator architecture centred on the ESP8266 NodeMCU. The overall data flow is: sensors → GPIO reads → decision logic → relay actuation + LCD update + serial logging. The system operates in a continuous polling loop with a 2-second delay between iterations, governed by the DHT11's 1 Hz maximum sampling rate.

The architecture distinguishes two independent sensing subsystems and two independent actuation subsystems:

- Presence subsystem: IR obstacle sensor → D6 (GPIO12) → occupancy state variable → Relay 1 (motion LEDs).
- Environmental subsystem: DHT11 → D5 (GPIO14) → temperature/humidity variables → Relay 2 (temperature LEDs, conditionally gated on occupancy state).
- Display subsystem: All state variables → 16×2 I2C LCD via D2 (SDA) / D1 (SCL).
- Communication subsystem: ESP8266WiFi + ESP8266WebServer → 802.11 b/g/n LAN → future HTTP endpoints.



Overall Data Flow in the System :

The complete system operates in the following sequence:

1. Sensors detect room conditions (presence, temperature, humidity).
2. Sensor signals are sent to the **ESP8266 NodeMCU**.
3. The controller processes the data using programmed logic.
4. Based on conditions:
   - Relays activate LEDs
   - LCD updates display information
5. Wi-Fi module enables monitoring through network interfaces.

3.2 COMPARISON OF SMART ROOM HUB WITH STANDARD SYSTEMS

Feature	Standard System	Smart Room Hub	Advantage
Motion Detection	PIR motion sensor	IR obstacle sensor	Continuous presence detection
Detection Method	Movement-based	Presence-based (continuous)	No false negatives for stationary users
Sensing Range	5–7 meters	2–30 cm (adjustable)	Fine-grained proximity control
Environmental Data	Temperature only	Temperature + Humidity	Comprehensive monitoring
Temp Sensor	DS18B20 (1-Wire)	DHT11 (Singlewire)	Humidity reading included
LCD Connection	Standard I2C	Custom I2C (D2SDA, D1-SCL)	Non-default pin flexibility
Logic Design	Independent control	Conditional hierarchy	Energy-efficient automation
WiFi Integration	Optional add-on	Built-in web server ready	Future-proof architecture

Table 1. Comparison of Smart Room Hub system with conventional home automation approaches.

3.3 HIERARCHICAL CONTROL LOGIC

The control logic implements a two-level hierarchy:

3.3.1 LEVEL 1 — OCCUPANCY DETECTION

The IR obstacle sensor output (D6) is polled each loop iteration. The active-LOW signal (LOW = person present) is read and stored in the Boolean variable personPresent. A timestamp-based timeout prevents false negatives:

```
bool detected = (digitalRead(IR_SENSOR) ==
LOW); if (detected) { personPresent = true;
lastSeenTime = millis();
} if (millis() - lastSeenTime >
presenceTimeout) { personPresent = false; }
```

The presenceTimeout constant (default 10,000 ms) is adjustable in source code to accommodate different detection scenarios. This mechanism allows a seated occupant whose movement ceases to remain registered as present for a configurable grace period.

3.3.2 LEVEL 2 — TEMPERATURE CONTROL (CONDITIONAL)

Temperature-based relay actuation is gated by the occupancy state, ensuring that temperature indicators remain inactive when the room is unoccupied:

```
if  (personPresent)  {  if  (tempC  >=  25)  {
    digitalWrite(RELAY4, LOW); // Temp GREEN ON
    digitalWrite(RELAY3, HIGH); // Temp RED OFF
    } else { digitalWrite(RELAY4, HIGH); // Temp
    GREEN OFF digitalWrite(RELAY3, LOW); // Temp
    RED ON }
} else { digitalWrite(RELAY4, HIGH); //
    Both OFF digitalWrite(RELAY3, HIGH);
}
```

3.4 GPIO PIN ASSIGNMENT

Function	NodeMCU Pin	GPIO#	Component	Notes
LCD I2C SCL	D1	5	LCD SCL	I2C clock line
LCD I2C SDA	D2	4	LCD SDA	I2C data line
Relay Motion RED	D3	0	Relay1 IN1	Motion absent indicator
Relay Motion GREEN	D4	2	Relay1 IN2	Motion present indicator
DHT11 Data	D5	14	DHT11 Data	Temp/humidity sensor
IR Sensor OUT	D6	12	IR Obstacle OUT	LOW = person detected
Relay Temp RED	D7	13	Relay2 IN1	Cold temp alert (<25°C)

Relay Temp GREEN	D0	16	Relay2 IN2	Comfortable temp (≥25°C)
Power Input	VIN	—	MB102 +5V	Microcontroller power
Ground	GND	—	All module GND	Common ground bus

Table 2. Complete GPIO pin mapping for the Smart Room Hub system.

3.5 LED FUNCTIONAL ASSIGNMENTS

LED Color	Relay Channel	Function	Active Condition
RED	Relay1-CH1 (D3)	Motion Absent	No person detected
GREEN	Relay1-CH2 (D4)	Motion Present	Person detected
RED	Relay2-CH1 (D7)	Cold Temp Alert	Temp < 25°C AND person present
GREEN	Relay2-CH2 (D0)	Comfortable Temp	Temp ≥ 25°C AND person present

Table 3. LED indicator color coding and relay channel assignments.

3.6 POWER DISTRIBUTION STRATEGY

All components are powered from the MB102 breadboard power supply module operating at 5V output. The NodeMCU receives 5V via its VIN pin and regulates it internally to 3.3V for its SoC. The relay module coils, IR sensor, DHT11, and LCD I2C backpack all operate at 5V. A 100 µF electrolytic capacitor is installed across the breadboard power rails to absorb switching transients generated by relay coil energization and de-energization, protecting the NodeMCU's internal voltage regulator.

A critical safety constraint: the NodeMCU must not be powered simultaneously from USB and the VIN pin with an external 5V supply connected, as this would forwardbias the onboard USB protection diode and potentially damage the voltage regulator.

CHAPTER 4

IMPLEMENTATION

4.1 COMPONENT SELECTION AND BILL OF MATERIALS

Component	Qty	Purpose	Specifications
ESP8266 NodeMCU	1	Main controller	3.3V logic, 11 GPIO, WiFi 802.11 b/g/n
MB102 Breadboard PSU	1	Power supply	5V/3.3V output, 800 mA max
100 µF Capacitor	1	Power stabilization	Reduces voltage noise
2-Channel 5V Relay (×2 modules)	2	Control 4 LED loads	Opto-isolated, Active LOW, 5V coil
IR Obstacle Sensor	1	Presence detection	3.3–5V, OUT: LOW = present, adjustable
DHT11 Sensor	1	Temp & humidity	0–50°C, 20–90% RH, ±2°C, ±5%
16×2 LCD + I2C Module	1	Status display	I2C address: 0x27 or 0x3F
5 mm RED LED	2	Alert indicators	20 mA, 2–2.2V forward voltage
5 mm GREEN LED	2	Normal indicators	20 mA, 2.8–3.2V forward voltage
220 Ω Resistor	4	LED current limiting	1/4W carbon film
830-point Breadboard	1	Component mounting	Standard 830 tie points
Jumper Cables	25–30	Wiring connections	Male-to-male assorted

Table 4. Complete bill of materials for the Smart Room Hub system.

The total estimated cost of the system is ₹1,800–₹2,800 INR (approximately USD 22– 34), making it accessible for student-level prototyping and educational deployment. Components were selected on the basis of availability in the Indian electronics retail market (e.g., Robu.in, Electronix Hub, and local component stores in Chennai).

4.2 CIRCUIT ASSEMBLY

4.2.1 POWER DISTRIBUTION SETUP

The MB102 module's jumper is set to the 5V position. The positive and negative output terminals are connected to the breadboard's red and blue power rails respectively. The 100 μF capacitor is installed across these rails with correct polarity (longer leg to positive rail). The NodeMCU's VIN pin connects to the positive rail and GND to the negative rail.

4.2.2 SENSOR CONNECTIONS

The IR obstacle sensor is wired with VCC to the 5V rail, GND to the negative rail, and the OUT pin to NodeMCU D6 (GPIO12). The detection range potentiometer on the sensor module is adjusted to the desired sensitivity before final assembly. The DHT11 sensor connects with VCC to 5V, GND to GND, and the data pin directly to NodeMCU D5 (GPIO14); no external pull-up resistor is required as the sensor module includes a built-in 10 kΩ pull-up resistor on the data line (ElectronicWings, 2019).

4.2.3 LCD I2C DISPLAY CONNECTIONS

The I2C backpack module is connected with VCC to 5V, GND to GND, SDA to NodeMCU D2 (GPIO4), and SCL to NodeMCU D1 (GPIO5). The firmware initializes the I2C bus with Wire.begin(D2, D1) to assign the non-default pin mapping. The I2C address 0x27 is used in the LiquidCrystal_I2C constructor; if the LCD is unresponsive, an I2C scanner sketch should be run to verify the actual address of the I2C backpack (0x3F is an alternative).

4.2.4 RELAY MODULE CONNECTIONS

Two 2-channel relay modules are employed. **Relay Module 1** (motion indicators): IN1

→ D3 (GPIO0) for motion RED; IN2 → D4 (GPIO2) for motion GREEN. **Relay Module 2** (temperature indicators): IN1 → D7 (GPIO13) for temperature RED; IN2 → D0 (GPIO16) for temperature GREEN. Both modules have VCC connected to the 5V rail and GND to the negative rail. The opto-isolation feature of each relay module ensures that the relay coil's switching noise is decoupled from the NodeMCU's logic domain.

4.2.5 LED INDICATOR CONNECTIONS

Each relay module's NO (normally open) contact serves as the switched 5V supply for its associated LED. Connection path: Breadboard positive rail → relay COM → relay NO → LED anode (+) → LED cathode (−) → 220 Ω resistor → breadboard negative rail. The 220 Ω resistor limits current to approximately 10–14 mA for both RED (Vf ≈
2.0 V) and GREEN (Vf ≈ 3.0 V) LEDs, well within their 20 mA maximum ratings.

4.3 SOFTWARE IMPLEMENTATION

4.3.1 LIBRARY DEPENDENCIES

The following libraries must be installed in Arduino IDE prior to compilation:

- ESP8266WiFi — included with the ESP8266 board package
- ESP8266WebServer — included with the ESP8266 board package
- DHT sensor library — by Adafruit (install via Library Manager)
- LiquidCrystal I2C — by Frank de Brabander (install via Library Manager)
- Wire — included with Arduino IDE core

The ESP8266 board package is added to Arduino IDE by appending the following URL to File → Preferences → Additional Board Manager URLs:

```
http://arduino.esp8266.com/stable/package_esp8266com_index.json
```

4.3.2 PIN AND CONSTANT DEFINITIONS

```
#define DHTPIN D5 // DHT11 data pin (GPIO14)
#define DHTTYPE DHT11
#define IR_SENSOR D6 // IR obstacle sensor OUT pin (GPIO12)
#define RELAY1   D3   // Motion RED indicator (GPIO0)
#define RELAY2   D4   // Motion GREEN indicator (GPIO2)
#define RELAY3   D7   // Temperature RED indicator (GPIO13)
#define RELAY4   D0   // Temperature GREEN indicator (GPIO16)
const unsigned long presenceTimeout = 10000; // 10 seconds
```

4.3.3 SETUP FUNCTION

The setup() function performs initialization in the following order:

6. Serial monitor initialized at 115200 baud.
7. I2C bus initialized with Wire.begin(D2, D1) for non-default pin assignment.

8. LCD initialized and startup message displayed for 2 seconds.
9. Relay GPIO pins configured as OUTPUT and set HIGH (all relays OFF).
10. DHT11 sensor initialized with dht.begin().
11. WiFi connection established; LCD displays IP address for 3 seconds.
12. Web server started with server.begin().

4.3.4 MAIN LOOP

The loop() function executes the following sequence each iteration:

13. Read IR sensor; update personPresent and lastSeenTime.
14. Evaluate 10-second timeout; set personPresent = false if elapsed.
15. Read temperature and humidity from DHT11.
16. Apply motion relay control (Relay 1 pair).
17. Apply conditional temperature relay control (Relay 2 pair).
18. Print status to Serial monitor.
19. Update LCD display with current readings.
20. Delay 2000 ms (to respect DHT11 sampling limit).

4.3.5 LCD DISPLAY FORMAT

The 16×2 display presents information in the following format:

```
Line 1: T:23.5C H:65% (temperature 1 dp, humidity integer)
Line 2: Person:Yes       (or Person:No )
```

4.4 UPLOAD PROCEDURE

21. Connect NodeMCU to computer via USB micro-B cable.
22. In Arduino IDE, select Tools → Board → NodeMCU 1.0 (ESP-12E Module).
23. Select the correct COM port under Tools → Port.
24. Update WiFi credentials in the ssid and password constants (lines 8–9 of the sketch).
25. Click the Upload button; wait for the "Hard resetting via RTS pin..." confirmation.

26. Open Serial Monitor at 115200 baud to verify WiFi connection and IP address.

4.5 ESP CODE

```
#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>
#include      <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// ---------------- WIFI ------------
---- const char* ssid =
"JOSHGREE7"; const char*
password = "patanahi";

ESP8266WebServer server(80);

// ---------------- PINS ----------------
#define DHTPIN  D5
#define DHTTYPE DHT11
#define IR_SENSOR D6
#define RELAY1 D3   // Motion RED
#define RELAY2 D4   // Motion GREEN
#define RELAY3 D7   // Temp RED
#define RELAY4 D0   // Temp GREEN

// ---------------- LCD ----------------
LiquidCrystal_I2C lcd(0x27,16,2);

// ---------------- SENSOR ----------------
DHT dht(DHTPIN, DHTTYPE);
```

```
// ---------------- VARIABLES ----------
------ bool personPresent = false; float
tempC = 0.0; float humidity = 0.0;
unsigned long lastSeenTime = 0;

const unsigned long presenceTimeout = 10000;

void setup() {
  Serial.begin(115200);

  // LCD   Wire.begin(D2,
D1);   lcd.begin(16,2);
lcd.backlight();
lcd.setCursor(0,0);
lcd.print("Smart Room
Hub");   delay(2000);
lcd.clear();

  pinMode(IR_SENSOR,
INPUT);   pinMode(RELAY1,
OUTPUT);
pinMode(RELAY2, OUTPUT);
pinMode(RELAY3, OUTPUT);
pinMode(RELAY4, OUTPUT);

  // Relays OFF initially (active
LOW)   digitalWrite(RELAY1,
HIGH);   digitalWrite(RELAY2,
HIGH);   digitalWrite(RELAY3,
HIGH);   digitalWrite(RELAY4,
HIGH);   dht.begin();
WiFi.begin(ssid, password);
lcd.print("Connecting WiFi");
```

```
  Serial.println("Connecting WiFi...");

  while (WiFi.status() != WL_CONNECTED)
 {
delay(50
0);
Serial.pr
int(".");
  }

  Serial.println("\nWiFi
Connected");
Serial.println(WiFi.localIP());
lcd.clear();   lcd.setCursor(0,0);


  lcd.print("WiFi
Connected");
lcd.setCursor(0,1);
lcd.print(WiFi.localIP());
delay(3000);   lcd.clear();
server.begin(); } void
loop() {


  // -------- IR SENSOR PRESENCE --------   bool
detected = (digitalRead(IR_SENSOR) == LOW);
if (detected) {     personPresent = true;
lastSeenTime = millis();
  }
  if   (millis()   -   lastSeenTime   >   presenceTimeout)   {
personPresent = false;
  }
```

```
  // -------- READ DHT --------
tempC = dht.readTemperature();
humidity = dht.readHumidity();
// -------- MOTION RELAYS --
------   if (personPresent) {
   digitalWrite(RELAY2,     LOW);             //     Motion     GREEN     ON
digitalWrite(RELAY1, HIGH);  // Motion RED OFF
  }
el
se
{
   digitalWrite(RELAY2,     HIGH);          //     Motion     GREEN     OFF
digitalWrite(RELAY1, LOW);   // Motion RED ON
  }



  // -------- TEMPERATURE RELAYS (ONLY WHEN PERSON PRESENT) ---
-----   if(personPresent){     if (tempC >= 25) {
     digitalWrite(RELAY4,     LOW);             //     Temp     GREEN     ON
digitalWrite(RELAY3, HIGH);  // Temp RED OFF
    }
els
e {
     digitalWrite(RELAY4,     HIGH);          //     Temp     GREEN     OFF
digitalWrite(RELAY3, LOW);   // Temp RED ON
    }


  }
els
e{
   // Nobody present → both
OFF     digitalWrite(RELAY4,
```

```
HIGH);
digitalWrite(RELAY3, HIGH);
  }

  // -------- SERIAL OUTPUT --------
  Serial.print("Temp: ");
  Serial.print(tempC);
  Serial.print(" C | Humidity: ");
  Serial.print(humidity);
  Serial.print(" % | Person: ");
  Serial.println(personPresent ? "Present" : "Not Present");

  // -------- LCD DISPLAY --
------   lcd.setCursor(0,0);
lcd.print("T:");
lcd.print(tempC);
lcd.print("C H:");
lcd.print(humidity);
lcd.print("% ");
lcd.setCursor(0,1);
lcd.print("Person:");
  lcd.print(personPresent   ?   "Yes   "   :   "No      ");
delay(2000);
}
```

CHAPTER 5

RESULTS AND DISCUSSION

5.1 FUNCTIONAL TESTING OVERVIEW

Following assembly and firmware upload, the system was subjected to a structured functional validation covering all four major subsystems: presence detection, environmental sensing, relay and LED actuation, and display output. Testing was conducted in an indoor laboratory environment at approximately 23–25°C ambient temperature and 55–65% relative humidity.

5.2 PRESENCE DETECTION SUBSYSTEM

5.2.1 IR SENSOR CALIBRATION

The IR obstacle sensor's distance potentiometer was adjusted to detect an object (simulating a seated occupant's hand or torso) at a range of 10–15 cm. The onboard LED of the sensor module confirmed detection visually before integration with the NodeMCU. Detection reliability was tested with ten consecutive insertion and removal cycles; the sensor correctly toggled the OUT pin between LOW (present) and HIGH
(absent) in all ten trials.

5.2.2 TIMEOUT MECHANISM VALIDATION

The 10-second presence timeout was validated by placing an object in the sensor field, observing the motion GREEN LED illuminating within one loop iteration (approximately 2 seconds), then removing the object and timing the LED transition back to RED. The transition occurred consistently at 10–12 seconds after removal (within one 2-second loop cycle of the 10 000 ms timeout threshold), confirming correct millis()-based timeout operation.

5.3 ENVIRONMENTAL SENSING SUBSYSTEM

DHT11 readings were cross-validated against a calibrated digital thermometer and hygrometer. Temperature readings exhibited a consistent offset of approximately +1.5°C relative to the reference instrument, within the sensor's ±2°C specified tolerance. Humidity readings were within ±4% of the reference, also within specification. Body-heat testing (placing a fingertip on the sensor body) produced temperature readings rising from approximately 24°C to 33°C over a 15-second period, confirming sensor responsiveness. Serial monitor output confirmed that no NaN (Not a Number) readings occurred during a one-hour continuous test, indicating stable wiring and reliable communication.

5.4 RELAY AND LED ACTUATION SUBSYSTEM

5.4.1 MOTION LED LOGIC

With no person detected: motion RED LED ON, motion GREEN LED OFF (GPIO D3 LOW, D4 HIGH). With person detected: motion GREEN LED ON, motion RED LED OFF (GPIO D4 LOW, D3 HIGH). Both transitions were accompanied by audible relay clicks and confirmed with a multimeter measuring 5V across the LED anode-cathode junction when active.

5.4.2 TEMPERATURE LED LOGIC

With room unoccupied: both temperature LEDs remained OFF throughout testing regardless of ambient temperature, confirming the conditional gating. With room occupied at 23.5°C (< 25°C threshold): temperature RED LED ON, temperature GREEN LED OFF. After gently heating the DHT11 sensor with a heat gun from approximately 1 meter distance, when the reading crossed 25.0°C the relays clicked and the temperature GREEN LED activated while RED deactivated. Cooling produced the reverse transition.

5.4.3 ACTIVE LOW RELAY BEHAVIOR

At power-on, all relay GPIO pins initialize HIGH, keeping all relays in the off state. This was confirmed by observing all LEDs remain off during the 5-second initialization and WiFi connection phase before the main loop begins. This fail-safe behavior is consistent with Horowitz and Hill (2015) recommendations for embedded relay control.

5.5 DISPLAY SUBSYSTEM

The 16×2 LCD successfully displayed startup messages, WiFi IP address, and live environmental data across all test conditions. The I2C address was confirmed as 0x27 using an I2C scanner sketch. The contrast potentiometer on the I2C backpack was adjusted for optimal readability under laboratory lighting. Display updates occurred every 2 seconds, synchronized with DHT11 reading intervals.

5.6 WIFI CONNECTIVITY

The NodeMCU successfully connected to the 2.4 GHz WiFi network (WPA2-PSK) within approximately 3–5 seconds of power-on. The assigned IP address was displayed on the LCD and confirmed via the router's DHCP client list. The ESP8266WebServer was initialized; while no HTTP routes were defined in the base implementation, server.handleClient() was called in each loop iteration, establishing the infrastructure for future endpoint registration.

5.7 SYSTEM STABILITY

The system operated continuously for two hours without spontaneous reboot, NaN sensor readings, LCD corruption, or relay mis-state. Power supply voltage remained at 5.05 V under full load (all four relays energized simultaneously), within the MB102 module's specification. The 100 µF decoupling capacitor successfully suppressed relayinduced voltage spikes that without it caused occasional NodeMCU brownout resets during testing.

5.8 COMPARISON WITH PROJECT OBJECTIVES

All six objectives defined in Section 1.3 were achieved. Continuous presence detection with a 10-second timeout was implemented and validated (Objective 1). Real-time temperature and humidity sensing with occupancy-conditional activation was confirmed operational (Objective 2). Four LED indicators with correct hierarchical relay logic were verified (Objective 3). LCD display updated at 2-second intervals with accurate data was demonstrated (Objective 4). WiFi web server framework was established (Objective 5). APA-formatted documentation and literature references are provided throughout this report (Objective 6).

5.9 TROUBLESHOOTING GUIDE

Issue	Possible Cause	Solution
LCD blank / no backlight	Wrong I2C address or wiring	Run I2C scanner sketch; check D1/D2 connections
LCD shows boxes	Contrast potentiometer too high	Rotate I2C backpack potentiometer counter-clockwise
Temp reads NaN	DHT11 not properly connected	Verify D5 wiring; check sensor module
IR always HIGH / LOW	Sensitivity mis-set or wiring issue	Adjust distance potentiometer; verify D6 connection
Relays do not click	Wrong GPIO or insufficient power	Verify pin numbers; confirm 5V supply
LEDs do not light	Polarity reversed or wrong resistor	Check anode/cathode orientation; verify 220 Ω resistors
WiFi will not connect	Wrong credentials or 5 GHz band	Update SSID/password in code; ensure router uses 2.4 GHz
Upload fails	Driver or board selection error	Install CH340 driver; select NodeMCU 1.0 (ESP-12E) board
Temp relays always OFF	Person not detected by IR sensor	Resolve IR sensor issue first (presence is prerequisite)
System reboots randomly	Insufficient power supply	Use 1A+ USB adapter; verify capacitor installation

Table 5. Troubleshooting guide for common Smart Room Hub assembly and runtime issues.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project successfully designed, implemented, and validated an IoT-based Smart Room Hub system using the ESP8266 NodeMCU microcontroller. The system addresses the three principal shortcomings of conventional entry-level automation systems: transient presence detection, independent sensing subsystems, and limited user feedback.

The key technical contributions of this work are:

- Continuous Presence Detection: The deployment of an IR obstacle sensor with a 10-second presence timeout delivers reliable occupancy monitoring for stationary occupants—a scenario where PIR sensors systematically fail.
- Hierarchical Energy-Efficient Control: Temperature indicators are conditionally activated only when room occupancy is confirmed, preventing unnecessary relay actuation and energy consumption in vacant rooms.
- Dual Environmental Parameter Monitoring: The DHT11 sensor provides simultaneous temperature and humidity data—a capability absent from most analogous student projects that rely on temperature-only DS18B20 sensors.
- Non-Default I2C Pin Assignment: The firmware correctly initializes the I2C bus on non-default NodeMCU pins (D2 = SDA, D1 = SCL) via Wire.begin(D2, D1), a configuration often overlooked in ESP8266 LCD integration tutorials.
- WiFi-Ready Architecture: The integrated ESP8266WebServer framework provides a ready-made extension point for HTTP-based remote monitoring without requiring hardware modifications.

The system operated without failure during a two-hour continuous test, meeting all six project objectives. The total component cost of ₹1,800–₹2,800 INR demonstrates the economic viability of IoT prototyping for educational and residential applications. The project provides a practical foundation for understanding embedded systems integration, sensor interfacing, relay control, I2C communication, and WiFi networking—core competencies in electronics and computer engineering curricula.

6.2 FUTURE WORK

Several directions for system enhancement are identified for future investigation:

6.2.1 WEB-BASED REMOTE MONITORING

HTTP GET endpoints (/data returning JSON, / returning an HTML dashboard) should be implemented using the existing ESP8266WebServer framework. This would enable real-time monitoring from any browser on the local network without additional hardware.

6.2.2 CLOUD INTEGRATION

ThingSpeak or Blynk integration would enable historical data logging and trend analysis. Temperature and humidity time series could reveal patterns in room usage and environmental variation, supporting data-driven HVAC scheduling.

6.2.3 ENHANCED SENSING

The addition of an LDR (light-dependent resistor) via the A0 analogue input would enable ambient light-based automation. An MQ-2 gas sensor would add safety monitoring capability (LPG, smoke). Replacing the IR sensor with an HC-SR04 ultrasonic sensor would extend detection range to 2–400 cm, making the system suitable for larger rooms.

6.2.4 MULTI-ROOM EXPANSION

A mesh of NodeMCU units, each acting as an MQTT publisher, could report to a central Raspberry Pi MQTT broker running Node-RED, creating a whole-house automation dashboard. This architecture scales to arbitrary room counts without modification to individual unit firmware.

6.2.5 RTC-BASED SCHEDULING

A DS3231 real-time clock module connected via I2C would enable time-aware automation: for example, disabling presence-based control during defined sleep hours, or applying different temperature thresholds for morning versus afternoon occupancy.

6.2.6 PCB MINIATURIZATION

The breadboard prototype should be transferred to a custom PCB for a permanent installation-grade deployment. KiCad or EasyEDA schematic capture and PCB layout tools are appropriate for this next stage, and the component footprints are all available in standard open-source libraries.

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